1. **Python program to find a side of a right-angled triangle whose two sides and an angle is given as base=2.3, height=6.7 and angle=60.**

2. **Program to Put Even and Odd elements in a List into Two Different Lists. Given lists are L= [2,19,24,56,77,59,90,62,46].**

3. **Python program to obtain the marks in computer of some students and find mean median and mode. The marks are given as 84,95,76,54,96,88,91,92,95,75,91,97,80**

4. **Program to Generate Random Numbers from 1 to 20 and append them to the List.**

5. **Python program to accept the base and perpendicular and find the hypotenuse of a right-angled triangle.**
6. **Program to Find all Numbers in a Range which are Perfect Squares and Sum of all Digits in the Number is Less than 10.**

7. **Write a Python program to read 2 numbers from user. Display the result using bitwise and, bitwise or, bitwise xor, right shift and left shift operator**
8. **Program to Remove the Duplicate Items from a List.**

9. **Write a python program to check the given date by the user is valid.**
10. **Program to concatenate dictionaries to create a new one.**

11. **Write a python program to find the roots of a quadratic equation. The standard form of quadratic equation is**

```
ax² + bx + c = 0, where
a, b and c are real numbers and
a ≠ 0
```

12. **Program to check whether a given key already exists in a dictionary.**

13. **Python Program to Convert Decimal to Binary, Octal and Hexadecimal.**
14. **Program to merge two Python dictionaries.**

15. **WAP to read a 4-digit number through the keyboard and calculate sum of digits.**

16. **Create a function named count vowels that accepts a string and returns the number of vowels in the string.**
17. **Program to check if a year is leap**
18. **Program to Find the Factorial of a Number Using Recursion.**
19. **Given a string. Delete from it all the characters whose indices are divisible by 3.**
20. **Program to Update the first set with items that don't exist in the second set. Consider any 2 sets.**

21. **1.Write a program to reverse 4-digit number using % and // operators.**
22. **2. Program to concatenate dictionaries to create a new dictionary.**
23. **WAP to calculate distance between 2 points. Distance formula is given as.**
    a. **Distance Formula =** $((x_2 - x_1)^2 + (y_2 - y_1)^2)^{½}$

24. **Python Program to calculate square and cube of a number and return Multiple Values from a Function.**
25. **WAP which prompts user to enter radius of circle. If the radius is greater than 0 and then calculate and print the area and circumference of the circle.**

26. **Function to check if a number is even or odd using return statement.**

27. **1.WAP to test whether a number is divisible by 3 and 15 or by 3 and 15.**

28. **2. Write a Python program that reads a text file, counts the number of words in it, and writes the word count to a new file**

29. **WAP to calculate the square of only those numbers whose least significant digit is 5.**

30. **Write a Python program that reads the content of one text file and writes it to another file.**
31. **WAP to calculate sum of digits of a number.**

32. **To keep record of patients' medical data, manipulate files to store, update, and delete such information.**
33. **WAP to calculate reverse of the number entered.**

34. **Function to Find the Sum of the Digits of the Number Recursively**

35. **WAP to check whether the number is Armstrong number or not.**

36. **Python Program to Compute the Power of a Number**

37. **WAP that prompts a user to enter four numbers and find the greatest number among the four numbers entered.**

38. **Python Program to Capitalize the First Character of a String.**

39. **WAP to display pattern of numbers as given:**
    i. **1**
    ii. **1 2**
    iii. **1 2 3**
    iv. **1 2 3 4**
    v. **1 2 3 4 5**

40. **Python Program to consider a List and print the Last Element of the List.**

41. **WAP to check number is prime or not using while loop.**

42. **Python Program to Swap Two Variables using user-defined functions.**

43. **Read the string "hello world" from the user. Make use of continue keyword and remove space.**

44. **Python Program to Display the multiplication Table using while loop.**

45. **A school decided to replace the desks in three classrooms. Each desk sits two students. Given the number of students in each class, print the smallest possible number of desks that can be purchased. The program should read three integers: the number of students in each of the three classes, a, b and c respectively. In the first test there are three groups. The first group has 20 students and thus needs 10 desks. The second group has 21 students, so they can get by with no fewer than 11 desks. 11 desks is also enough for the third group of 22 students. So we need 32 desks in total.**

46. **Python Program to Calculate the Area of a Triangle .**

47. **H hours, M minutes and S seconds are passed since the midnight ($0 \le H < 12$, $0 \le M < 60$, $0 \le S < 60$). Determine the angle (in degrees) of the hour hand on the clock face right now.**

48. **Python Program to Convert Celsius to Fahrenheit.**

49. **Given integer coordinates of three vertices of a rectangle whose sides are parallel to the coordinate axes, find the coordinates of the fourth vertex of the rectangle.**

In the first test the three given vertices are (1, 4), (1, 6), (7, 4). The fourth vertex is thus (7, 6).

50. Python Program to Find the Largest Among Three Numbers.

51. There was a set of cards with numbers from 1 to N. One of the card is now lost. Determine the number on that lost card given the numbers for the remaining cards. Given a number N, followed by N − 1 integers - representing the numbers on the remaining cards(distinct integers in the range from 1 to N). Find and print the number on the lost card.

52. Python Program to Find the Sum of Natural Numbers.

53. In bowling, the player starts with 10 pins at the far end of a lane. The object is to knock all the pins down. For this exercise, the number of pins and balls will vary. Given the number of pins N and then the number of balls K to be rolled, followed by K pairs of numbers (one for each ball rolled), determine which pins remain standing after all the balls have been rolled. The balls are numbered from 1 to N (inclusive) for this situation. The subsequent number pairs, one for each K represent the start to stop (inclusive) positions of the pins that were knocked down with each role. Print a sequence of N characters, where "I" represents a pin left standing and "." represents a pin knocked down.

54. Python Program to Find Numbers Divisible by Another Number.

55. A timestamp is three numbers: a number of hours, minutes and seconds. Given two timestamps, calculate how many seconds is between them. The moment of the first timestamp occurred before the moment of the second timestamp.

56. Python Program to Check if a Number is Positive, Negative or 0

57. WAP that prompts a user to enter 4 numbers and find the greatest number among the four numbers entered using for loop.

58. Python Program to Find ASCII Value of Character

59. WAP which prints the sum of every fifth number from 0 to 500(including 0 and 500) using while loop.

60. Python Program to Find LCM.

61. WAP to read password from user. If the user types the correct password i.e "python" then display message "welcome to python programming".

Note:only 3 attempts are allowed to enter correct password.

62. Python Program to Find Sum of Natural Numbers Using Recursion

63. WAP to read the positive integer and count the number of decimal digits in a positive number.

64. Python Program to Add Two Matrices.

65. Program to display first n Fibonacci numbers

66. Python Program to Access Index of a List Using for Loop

67. WAP to check whether the accepted number is palindrome or not.

68. Python Program to Get a Substring of a String

69. WAP to print following pattern:

i. **A**
ii. **B C**
iii. **D E F**
iv. **G H I J**
v. **K L M N O**

70. Python Program to Count the Occurrence of an Item in a List.

71. Consider a library management system. Each book is represented as a key in a dictionary, and its value is another dictionary containing information such as author, genre, and availability status.

a. **Write a function to add new books to the library.**
b. **Update information about existing ones.**
c. **Develop a function to search for books by a specific author.**

72. Write a python program to find roots of quadratic equation.

73. Consider a database for movies. The database is a dictionary where the keys are movie titles, and the values are dictionaries containing information such as genre, year, and rating.

a. **Write a function to add a new movie to the database or update information about an existing movie.**
b. **Create a function to list all movies of a specific genre.**
c. **Delete movie released before a specified year.**

74. Write a python program to display the multiplication table using a while loop.

75. **Consider building a system to track student grades. Each student is identified by their name (key), and their grades are stored as a list of numbers (value).**

    a. **Write a function to add a new student to the system or update an existing student's grades.**
    b. **Create a function to calculate and display the average grade for each student.**
    c. **Implement a function to find the student(s) with the highest average grade.**

76. **Write a python program to check if a year is leap or not.**

77. **Given a string. Delete from it all the characters whose indices are divisible by 3**

78. **Write a program to concatenate dictionaries to create a new one.**

79. **A timestamp is three numbers: a number of hours, minutes and seconds. Given two timestamps,calculate how many seconds are between them. The moment of the first timestamp occurred before the moment of the second timestamp.**

80. **Write a program to count the vowels in any given string**